

Dataset Processing Report

- Thesis Title: Impact of Speed Violation on Crash Severity.
- This report shows how the dataset was cleaned and prepared for Machine Learning.
- Raw Dataset → Clean Dataset → Machine Learning Model

Dataset Overview

- 10,000 Accidents
- 46 Original Columns
- such as Location, Time, Weather, Road, Severity
- Not a single column has been deleted.
- Only some new features have been added.
- So 46 Columns.
- Feature Engineering = 144 Columns

Target Variable

- Earlier, it was Severity
- This means how serious the accident is.
- Its value was 1, 2, 3, 4
- But later to make Machine Learning easier, it was changed to
- Low Severity and
- High Severity
- That is, 1 and 2 = Low, 3 and 4 = High

Data Validation

- Here only checked
- ✓ Are there any duplicates?
- ✓ Is the ID the same?

- ✓ Is the Data Type correct?
- Result
- No duplicates found.
- That is 10,000 Accidents = 10,000 different Accidents
- Time has been converted from Text to DateTime.
- So that Hour Month Year can be easily extracted.

Missing Value Handling

- The dataset did not have much information, such as Wind Speed, Temperature, Humidity, etc.
- Is the column being deleted? No. It was not deleted.
- So what did they do?
- Where there is no information, they filled it in with Median Mode or 0
- This is called Imputation i.e., filling in the blanks with reasonable values.

Outlier Treatment

- Outlier means an unusual value. Example All cars have a speed 50, 60, 55, 58
- But one is 800 km/h
- This is an Outlier.
- These have not been deleted.
- They have only been brought within the Limit.
- This is called Winsorization.
- That is, a very large value is reduced a little.

Datatype Correction

- New Information is Extracted from time
- For Example Hour, Month, Day, Night, Rush Hour, Weekend

- These help Machine Learning learn better.

Feature Engineering

- This is the most important part. Feature Engineering means creating new columns using old data.
- Suppose there is a Start time and an End time, then a new column can be created for Duration. This is Feature Engineering.
- A total of 22 new features have been created in the report.
- Road Feature
 - From road , its found out Highway, Interstate, State Road
- Speed Violation Risk
 - There is no column name speed violation in the Dataset
 - They created a score .
 - They assumed if the road is Highway, time is Night, Visibility is less and there is no traffic control then there is more risk of speed violation
 - A speed violation score is created from combining these information
- Weather Feature
 - The weather created a score called the Weather Severity Index, which measures how bad the weather was.
- Text Feature
 - Extracted from the Description Column how many letters, how many words. These are also added as new features.

Correlation Analysis

- Correlation means the relationship between two things. For example If speed increases, Accident Severity also increases then correlation is Positive.
- The report showed that Speed_Violation_Risk_Score had the highest relationship with crash severity.
- This means that higher speed-related risk leads to more severe crashes.

Multicollinearity (VIF)

- This analysis checked whether two features provide similar information.
- For example, Temperature and Wind Chill contain almost the same information.
- However, both features were kept because the supervisor requested that the original columns should not be removed.

Encoding

- Machine learning models cannot understand text data directly.
- So, text values like Dhaka, Chittagong, and Rajshahi were converted into numbers.
- This process is called Encoding.

Feature Scaling

- All numerical features were scaled to a similar range.
- For example, values like 5 and 5000 can affect model performance if left unchanged.
- Therefore, all numerical values were brought to a similar scale using feature scaling.

Class Imbalance

- Initially, the Severity classes were imbalanced.
- Classes 2 and 3 had many more samples, while Classes 1 and 4 had very few.
- To improve model learning, the target was grouped into two classes: Low and High.

Train-Test Split

- The 10,000 records were divided into three parts: 70% Training (7,000), 15% Validation (1,500), and 15% Testing (1,500).
- The Training set was used to train the model, and the Validation set was used to evaluate and tune its performance.
- The Testing set was used for the final evaluation of the trained model.